

ZHANHAO (NEO) ZHANG

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Reinforcement Learning | Stochastic Modeling | Optimization

EDUCATION

Cornell University | Ph.D. in Operations Research

Expected May 2027

Advisors: Jim Dai and Manxi Wu

Columbia University | M.A. in Statistics

Feb 2022

University of Washington | B.S. in Computer Science; B.S. in Statistics

Jun 2020

INDUSTRY EXPERIENCE

Meta | Research Intern, Machine Learning (PhD)

May 2026 - Aug 2026

- Defined the research approach for an open-ended pilot project to select truthful supplemental campaign-performance signals for third-party AI advisers allocating advertiser budgets across publishers.
- Built a market simulator for counterfactual scenarios using literature-based transition equations with parameters calibrated to campaign data; compared simulated and observed trajectory distributions using KS and moment tests. Modeled unknown competitor performance using scenarios of Meta's performance relative to other publishers.
- Built an agentic system that uses multi-round adversarial debate among LLM agents role-playing domain experts to design interpretable decision graphs; trained graph parameters with LLM-guided updates and policy-gradient RL for deterministic, low-latency serving.
- Reduced advertiser budget cuts or increased allocations to Meta in **20%** of evaluated simulation scenarios compared with a no-steering baseline.
- Accelerated simulation by **~6x** and cut model-serving costs to **~1/20** of baseline through parallelization and caching; the team adopted the simulator and steering-policy outputs for **production integration**.

Akamai Technologies | Research Intern, Mapping

Jun 2025 - Aug 2025

- Developed a stochastic CDN traffic simulator to assess machine utilization and tail latency under candidate DNS load-balancing policies, modeling connection arrivals, request timing, and joint distributions of request counts, workloads, and connection lifetimes.
- Characterized five traffic categories from sampled production logs using connection-level features and k-means clustering; analyzed how traffic patterns and extreme workloads affect simulated machine utilization.
- Found agreement between simulated and observed mean machine utilization; evaluated existing load-balancing policies to establish baselines for future policy comparisons.

Aetna at CVS Health | Data Scientist

Jun 2021 - Jul 2022

- Built a distributed Monte Carlo simulation pipeline to estimate member disenrollment, completing **100K+** simulations on datasets with millions of rows within one hour through parallelization and vectorization.
- Developed optimization and machine-learning methods for experimental design of campaigns, risk-cohort identification, and churn analysis using quadratic programming, Dask, Hive, and supervised learning.

SELECTED RESEARCH

Atomic Proximal Policy Optimization for Electric Robo-Taxi Dispatch and Charger Allocation

With Jim Dai and Manxi Wu | Transportation Science: minor revision

A scalable reinforcement learning algorithm addresses dispatch and charging in an average-reward Markov decision process (MDP), with benchmark comparisons using NYC trip records.

Optimal Batched Scheduling of Stochastic Processing Networks Using Atomic Action Decomposition

With Jim Dai and Manxi Wu | Management Science: major revision, revision submitted

Atomic action decomposition of the scheduling MDP preserves the optimal long-run average reward while keeping the policy dimension independent of the number of servers.

Deep Learning Method for Stationary Distribution of Reflected Brownian Motion

With Jim Dai | Submitted to Stochastic Systems

A deep learning method learns Laplace transforms to estimate stationary tail probabilities in high dimensions.

TECHNICAL SKILLS

Methods: Reinforcement learning, LLM agents, optimization, stochastic modeling, simulation

Languages: Python, SQL, C/C++, R, Bash

Tools: PyTorch, NumPy, SciPy, scikit-learn, Pandas, PySpark, Dask, Hive, Joblib, Git

HONORS AND TEACHING

INFORMS TSL Data-Driven Research Challenge finalist (2025) | Cornell ORIE Outstanding Teaching Assistant Award (2026)
Instructor, Big Data Technologies, Cornell University (Spring 2024, 2025, 2026)